

Mercury Computer Systems Selects Avnet Design Services' ASIC Engineering Service to Accelerate Time-to-Market

Augments internal engineering team to add ASIC engineering expertise

Mercury Computer Systems is the leading independent producer of high-performance stream computing systems. These embedded computer systems transform streams

of data to visual information in real time for analysis and interpretation. Mercury's products play a critical role in a wide range of defense and medical imaging applications. In air, sea and land-based military platforms, these systems process real-time radar, sonar, and signal intelligence data. Mercury's systems are also used in state-of-the-art medical diagnostic devices, including magnetic resonance imaging (MRI), computed tomography (CT), and the rapidly growing field of digital X-ray.

"When Mercury Computer first began working with Avnet Design Services, back in 1992, we were a privately held \$26M company, strapped for engineering resources," explained George Koslosky, Director of Materials. "Today we are a public company with over \$100M in sales, but we still use Avnet Design Services to help with our LSI Logic ASIC designs. Utilizing the outsourcing services of Avnet Design Services allows us to focus our internal engineering resources where they add the most value. For example, we recently used Avnet Design Services on the design of our RACE++™ Series 400-MHz PowerPC 7400 Daughtercards."

The Avnet Design Service (ADS) ASIC engineering service gave Mercury Computer a variety of engineering skills to help get their ASIC designs into production quickly. "Working with Avnet Design Services allows

us to apply their knowledge and experience to the implementation phase of the project. Since we don't need to allocate internal engineering resources

for these functions, we can apply our talents to our most important areas architecture and system design issues," explained Mr. Koslosky. "The quality of Avnet Design Services people is outstanding. We have developed a great relationship and rapport. We think of Avnet Design Services as 'our engineers.' We count on them to guide us through the technology options and provide risk and time to market analysis, as well as quick access to LSI Logic resources, when required."

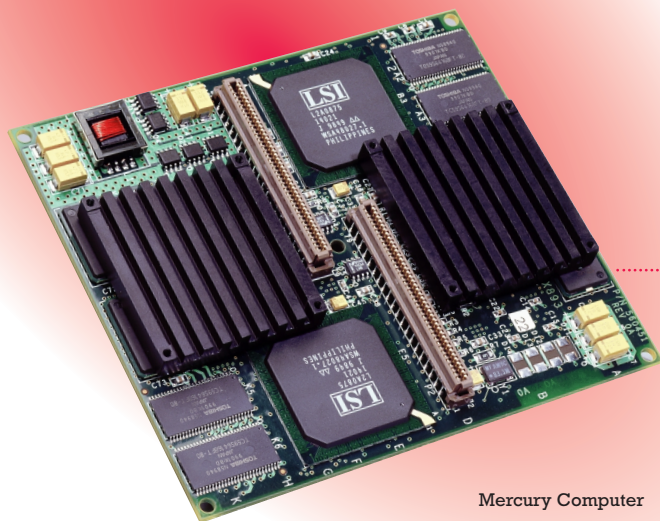
Mercury Computer not only used Avnet Design Services to shorten their ASIC designs time, they also relied on the extended Avnet organization to get the designs into production faster. Mr. Koslosky elaborated, "Avnet Design Services works closely with the Avnet Marshall part of the Avnet organization to ensure a smooth transition into production. They coordinate purchasing, order placement,

forecasting, bonding and a variety of other value-added services we have come to count on from Avnet in order to successfully hit our compressed time-to-market windows."

With the Mercury Computer RACE++ Series PowerPC 7400 daughtercard, embedded computing reaches a new level of performance. Each daughtercard contains two 400-MHz MPC7400 microprocessors with AltiVec™ technology. These unique microprocessors combine a modern superscalar RISC architecture with an AltiVec parallel vector execution unit. Each compute node on the daughtercard consists of the MPC7400 microprocessor, level 2 cache, synchronous DRAM and a special purpose ASIC. The ASIC contains architectural advancements that enhance concurrency between arithmetic and I/O operations. Many compute nodes can be utilized simultaneously by compute-intensive algorithms to support real-time image processing or 'visualization applications' that were previously impractical.

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*George Koslosky
Director of Materials
Mercury Computer Systems*



Mercury Computer
Race++ Series Daughtercard

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